

# MARIO FERNÁNDEZ PENDÁS

## PERSONAL INFORMATION

|                      |                                                              |
|----------------------|--------------------------------------------------------------|
| <i>date of birth</i> | 7 October 1988                                               |
| <i>nationality</i>   | Spanish                                                      |
| <i>email</i>         | <a href="mailto:mariofp77@gmail.com">mariofp77@gmail.com</a> |
| <i>phone</i>         | +34 628 650 863                                              |

## EDUCATION

|                                                       |                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>PhD in<br/>Mathematics and<br/>Statistics</i>      | <i>2014-2018</i> University of the Basque Country<br>Thesis: <i>Adapting Hybrid Monte Carlo methods for solving complex problems in Life and Materials sciences</i><br>Grade: <i>Sobresaliente cum laude</i> (highest grade)<br>Advisor: <a href="#">Prof. Elena AKHMATSKAYA</a> |
| <i>Masters in<br/>Mathematics and<br/>Computation</i> | <i>2011-2012</i> University of Cantabria<br>Thesis: <i>Some Metric Aspects in Algebraic Geometry, on the average</i><br>Advisor: <a href="#">Prof. Luis M. PARDO</a>                                                                                                             |
| <i>Bachelor in<br/>Mathematics</i>                    | <i>2006-2011</i> University of Oviedo<br>Thesis: <i>An approach to the P vs NP problem</i><br>Advisor: <a href="#">Prof. Elías FERNÁNDEZ-COMBARRO</a>                                                                                                                            |

## EMPLOYMENT

|                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>University of the<br/>Basque Country</i>                                      | <i>current</i> Assistant Professor, UNIVERSITY OF THE BASQUE COUNTRY<br>Assistant Professor in the UNIVERSITY OF THE BASQUE COUNTRY. Department of Quantitative Methods, Faculty of Economics and Business.                                                                                                                                                                                                                                                                                                                                                                            |
| <i>BCMaterials</i>                                                               | <i>2023-2025</i> Postdoc Fellow, BCMATERIALS<br>Postdoctoral Fellow in BCMATERIALS under the supervision of Professor Ivan Infante.<br>Reference: <a href="#">Prof. Ivan INFANTE</a>                                                                                                                                                                                                                                                                                                                                                                                                   |
| <i>University of the<br/>Basque Country,<br/>DIPC and Rutgers<br/>University</i> | <i>2020-2023</i> Postdoc Fellow (Joint Position), UNIVERSITY OF THE BASQUE COUNTRY and DONOSTIA INTERNATIONAL PHYSICS CENTER (DIPC), and RUTGERS UNIVERSITY<br>Postdoctoral Fellow in UNIVERSITY OF THE BASQUE COUNTRY and DONOSTIA INTERNATIONAL PHYSICS CENTER (DIPC) under the supervision of Professor Xabier López and Doctor David de Sancho.<br>Postdoctoral Fellow in RUTGERS UNIVERSITY under the supervision of Professor Darrin M. York.<br>References: <a href="#">Prof. Xabier LÓPEZ</a> , <a href="#">Dr. David DE SANCHO</a> , and <a href="#">Prof. Darrin M. YORK</a> |
| <i>BCAM</i>                                                                      | <i>2019</i> Visiting Postdoc Fellow, BASQUE CENTER FOR APPLIED MATHEMATICS (BCAM)<br>Postdoctoral Fellow in the MSLMS group under the supervision of Professor Elena Akhmatskaya.<br>Reference: <a href="#">Prof. Elena AKHMATSKAYA</a>                                                                                                                                                                                                                                                                                                                                                |

University of  
Oviedo

2018-2019 Assistant Professor of Applied Mathematics,  
UNIVERSITY OF OVIEDO

Assistant Professor of Applied Mathematics in the Department of Mathematics.  
Reference: **Prof. Jesús SUÁREZ PÉREZ DEL RÍO**

BCAM

2018 Postdoc Fellow, BASQUE CENTER FOR APPLIED  
MATHEMATICS (BCAM)

Postdoctoral Fellow in the MSLMS group under the supervision of Professor  
Elena Akhmatskaya.

Reference: **Prof. Elena AKHMATSKAYA**

BCAM

2014-2018 PhD Student, BASQUE CENTER FOR APPLIED  
MATHEMATICS (BCAM)

PhD student in the MSLMS group under the supervision of Professor Elena  
Akhmatskaya.

Reference: **Prof. Elena AKHMATSKAYA**

BCAM

2013-2014 Internship Student, BASQUE CENTER FOR APPLIED  
MATHEMATICS (BCAM)

Internship in the MSLMS group under the supervision of Professor Elena  
Akhmatskaya and Dr. Bruno Escribano.

Reference: PhD Bruno ESCRIBANO

## PUBLICATIONS

*Journal of  
Computational  
Physics*, 502,  
112800

2024 Adaptive multi-stage integration schemes for Hamiltonian  
Monte Carlo

Authors: Lorenzo NAGAR, Mario **Fernández-Pendás**, Jesús María SANZ-SERNA,  
Elena AKHMATSKAYA

Data from year 2023. Impact factor and total citations: 3.8; 65,825  
Category, position and quartile: Physics, Mathematical; 3/60; Q1

*Journal of  
Chemical  
Information and  
Modeling*, 62, 23,  
6069-6083

2022 AMBER Drug Discovery Boost tools: Automated workflow  
for production free energy simulation setup and analysis  
(ProFESSA)

Authors: Abir GANGULY, Hsu-Chun TSAI, Mario **Fernández-Pendás**, Tai-Sung  
LEE, Timothy J. GIESE, Darrin M. YORK

Impact factor and total citations: 5.6; 31,925  
Category, position and quartile: Chemistry, Medical; 11/60; Q1

*Journal of  
Computational and  
Applied  
Mathematics*, 404,  
113388

2022 A study of the performance of classical minimizers in the  
Quantum Approximate Optimization Algorithm

Authors: Mario **Fernández-Pendás**, Elías F. COMBARRO, Sofia VALLECORSA, José  
RANILLA, Ignacio F. RÚA

Impact factor and total citations: 2.4; 20,102  
Category, position and quartile: Applied Mathematics; 45/267; Q1

*Mathematics in  
Industry Reports*

2021 Synthesis of Monodisperse Spherical Nanocrystals

Authors: Vincent CREGAN, Revathi BACSA, Marc CALVO, Mario  
**Fernández-Pendás**, Brendan FLORIO, Antonio MARQUINA, Iain MOYLES, Helena  
RIBERA PONSÁ, Simone RUSCONI, Susana SERNA

*Progress in  
Industrial  
Mathematics:  
Success Stories*, 5,  
69

## 2021 Multiscale Modelling and Simulation of Advanced Battery Materials

Authors: Mauricio R. BONILLA, Fabián A. GARCÍA DAZA, Mario **Fernández-Pendás**, Javier CARRASCO, Elena AKHMATSKAYA  
Book chapter. Eds.: Manuel CRUZ, Carlos PARÉS, Peregrina QUINTELA

*Journal of  
Computational  
Physics*, 373,  
900-916

## 2018 Multi-stage splitting integrators for sampling with modified Hamiltonian Monte Carlo methods

Authors: Tijana RADIVOJEVIĆ, Mario **Fernández-Pendás**, Jesús María SANZ-SERNA, Elena AKHMATSKAYA

Impact factor and total citations: 2.845; 51,597  
Category, position and quartile: Mathematical Physics; 4/55; Q1

*Langmuir*, 33(42),  
11530-11542

## 2017 Adaptive Splitting Integrators for Enhancing Sampling Efficiency of Modified Hamiltonian Monte Carlo Methods in Molecular Simulation

Authors (in alphabetic order): Elena AKHMATSKAYA, Mario **Fernández-Pendás**, Tijana RADIVOJEVIĆ, Jesús María SANZ-SERNA

Impact factor and total citations: 3.789; 118,255  
Category, position and quartile: Physical Chemistry; 48/147; Q2.  
Multidisciplinary Chemistry; 50/171; Q2. Multidisciplinary Materials Science; 61/285; Q1

*Theoretical  
Chemistry  
Accounts*, 136(43),  
43

## 2017 Enhancing sampling in atomistic simulations of solid state materials for batteries: a focus on olivine NaFePO<sub>4</sub>

Authors: Bruno ESCRIBANO, Ariel LOZANO, Tijana RADIVOJEVIĆ, Mario **Fernández-Pendás**, Javier CARRASCO, Elena AKHMATSKAYA

Impact factor and total citations: 1.545; 6,961  
Category, position and quartile: Physical Chemistry; 106/147; Q3

*Journal of  
Computational  
Physics*, 327,  
434-449

## 2016 Adaptive multi-stage integrators for optimal energy conservation in molecular simulations

Authors: Mario **Fernández-Pendás**, Elena AKHMATSKAYA, Jesús María SANZ-SERNA

Impact factor and total citations: 2.746; 42,429  
Category, position and quartile: Mathematical Physics; 3/55; Q1. Computer Science, Interdisciplinary Applications; 26/105; Q1

*Journal of  
Molecular  
Modeling*, 20,  
2487

## 2014 Constant pressure hybrid Monte Carlo simulations in GROMACS

Authors: Mario **Fernández-Pendás**, Bruno ESCRIBANO, Tijana RADIVOJEVIĆ, Elena AKHMATSKAYA

Impact factor and total citations: 1.736; 4,463  
Category, position and quartile: Multidisciplinary Chemistry; 66/157; Q2.  
Computer Science, Interdisciplinary Applications; 34/102; Q2

*Confidential*

## 2014 Technical Report for the industrial project led by Math-in

Description: Math-in collaborative project "Demanda de sampling y optimización".

Authors: Tijana RADIVOJEVIĆ, Mario **Fernández-Pendás**, Elena AKHMATSKAYA

*Journal of  
Complexity*, 29,  
323-350

## 2013 An Arithmetic Poisson Formula for the multi-variate resultant

Authors (in alphabetic order): Mario **Fernández-Pendás**, Luis M. PARDO

Impact factor and total citations: 1.191; 723

Category, position and quartile: Applied Mathematics; 61/251; Q1

The data above was taken from the Journal Citation Reports. Data corresponds to the year of publication of each paper.

### INVITED PRESENTATIONS

- |                                                                           |                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>BCAM, Bilbao<br/>(Spain)</i>                                           | <p><i>May 3, 2023</i>      Markov State Models for understanding long-timescale dynamics of molecular systems</p> <p>BCAM Alumni &amp; Former Members Seminar.<br/>Authors: Mario <b>Fernández-Pendás</b></p>                                                                                                    |
| <i>DIPC, Donostia<br/>(Spain)</i>                                         | <p><i>July 12, 2019</i>      Adapting Hybrid Monte Carlo methods for solving complex physical problems</p> <p>Seminar on invitation of Prof. Xabier López.<br/>Authors: Mario <b>Fernández-Pendás</b></p>                                                                                                        |
| <i>Department of<br/>Mathematics,<br/>Shanghai<br/>University (China)</i> | <p><i>July 8, 2019</i>      Fail-Safe Hamiltonian Monte Carlo.</p> <p>Seminar on invitation of Prof. P. Zhu.<br/>Elena <b>Akhmatskaya</b>, Mario FERNÁNDEZ-PENDÁS, Felix MÜLLER, Jorge PÉREZ-HEREDIA, Tijana RADIVOJEVIĆ, J.M. SANZ-SERNA</p>                                                                    |
| <i>CRM, Barcelona<br/>(Spain)</i>                                         | <p><i>June 7, 2019</i>      Fail-Safe Hamiltonian Monte Carlo</p> <p>Seminar on invitation of Prof T. Alarcón.<br/>Authors: Elena <b>Akhmatskaya</b>, Mario FERNÁNDEZ-PENDÁS, Felix MÜLLER, Jorge PÉREZ-HEREDIA, Tijana RADIVOJEVIĆ, J.M. SANZ-SERNA</p>                                                         |
| <i>BCAM, Bilbao<br/>(Spain)</i>                                           | <p><i>April 12, 2018</i>      Hybrid Monte Carlo methods for sampling in complex physical systems</p> <p>LIGHT PhD Seminar.<br/>Authors: Mario <b>Fernández-Pendás</b></p>                                                                                                                                       |
| <i>University of Bath,<br/>Bath (United<br/>Kingdom)</i>                  | <p><i>September 11, 2017</i> Adaptive splitting integrators for modified Hamiltonian Monte Carlo methods</p> <p>SciCADE Bath 2017.<br/>Authors: Elena <b>Akhmatskaya</b>, Mario FERNÁNDEZ-PENDÁS, Tijana RADIVOJEVIĆ, Jesús María SANZ-SERNA</p>                                                                 |
| <i>University of<br/>Valladolid,<br/>Valladolid (Spain)</i>               | <p><i>January 26, 2017</i> Adaptive splitting integrators for enhancing sampling efficiency of shadow Hamiltonian Monte Carlo methods</p> <p>IMUVA Seminar on invitation of Prof. J.M. Sanz-Serna.<br/>Authors: Mario <b>Fernández-Pendás</b>, Tijana RADIVOJEVIĆ, Jesús María SANZ-SERNA, Elena AKHMATSKAYA</p> |

ICERM, Brown  
University,  
Providence (USA)

July 21, 2016 Adaptive two-stage integrators for sampling  
algorithms based on Hamiltonian dynamics

Stochastic numerical algorithms, multiscale modeling and high-dimensional  
data analytics.

Authors: Elena **Akhmatskaya**, Mario FERNÁNDEZ-PENDÁS, Tijana RADIVOJEVIĆ,  
Jesús María SANZ-SERNA

University of  
Potsdam, Potsdam  
(Germany)

September 15, 2015 Adaptive multi-stage integrators with optimal  
energy conservation

SciCADE Potsdam 2015.

Authors: Mario **Fernández-Pendás**, Elena AKHMATSKAYA, Jesús María  
SANZ-SERNA

ITQB, Lisboa  
(Portugal)

December 4, 2014 Enhanced Sampling hybrid Monte Carlo  
Methods in GROMACS Package: MultiHMC-GROMACS

Seminar on invitation of Prof. A.M. Baptista.

Authors: Elena **Akhmatskaya**, Bruno ESCRIBANO, Mario FERNÁNDEZ-PENDÁS,  
Ivan TERTEROV

Repsol  
Technological  
Center, Madrid  
(Spain)

June 25, 2014 Improving the Efficiency in Sampling of the  
Posterior PDF for Reservoir-model Parameters

Technological meeting for the analysis of the Brugge model using mathematical  
technology.

Authors: Tijana **Radivojević**, Mario **Fernández-Pendás**, Elena AKHMATSKAYA

University of  
Valladolid,  
Valladolid (Spain)

January 22, 2014 Momentum Flips in Generalized  
Hybrid/Hamiltonian Monte Carlo Methods

Seminar on invitation of Prof. J.M. Sanz-Serna.

Authors: Tijana **Radivojević**, Elena AKHMATSKAYA, Bruno ESCRIBANO, Mario  
FERNÁNDEZ-PENDÁS

BCAM, Bilbao  
(Spain)

October 17, 2013 Enabling Constant Pressure Hybrid Monte  
Carlo Simulations Using the GROMACS Molecular Simulation  
Package

The Second BCAM Workshop on Computational Mathematics.

Authors: Mario **Fernández-Pendás**, Bruno ESCRIBANO, Elena AKHMATSKAYA

#### CONTRIBUTED PRESENTATIONS

AI4AM 2025,  
Donostia (Spain)

April 8, 2025 Implementation of a Machine Learning Force  
Fields Platform for Quantum Dots (poster)

AI4AM 2025 Artificial Intelligence for Advanced Materials Conference.

Authors: Zain Ul **Abideen**, Mario FERNÁNDEZ-PENDÁS, Ivan INFANTE

AI4AM 2025,  
Donostia (Spain)

April 8, 2025 Generating Machine Learning Force Fields for  
Colloidal Quantum Dots. The Case for CdSe, PbSe and InP  
(poster)

AI4AM 2025 Artificial Intelligence for Advanced Materials Conference.

Authors: Muhammad **Usman**, Zain Ul ABIDEEN, Mario FERNÁNDEZ-PENDÁS,  
Ivan INFANTE

MATSUS  
Conference,  
Barcelona (Spain)

March 6, 2024 Machine Learning Force Fields for Colloidal  
Quantum Dots

MATSUS Conference.

Authors: Mario **Fernández-Pendás**, Ivan INFANTE

- DIPC, Donostia (Spain)*      *September 29, 2016* Adaptive multi-stage integrators for optimal energy conservation in molecular simulations (poster)  
DYNAPEUTICS.  
Authors: Mario **Fernández-Pendás**, Elena AKHMATSKAYA, Jesús María SANZ-SERNA
- CECAM-HQ-EPFL, Lausanne (Switzerland)*      *February 15, 2016* Adaptive multi-stage integrators for optimal energy conservation in molecular simulations (poster)  
CECAM workshop on Models for Protein Dynamics 1976-2016.  
Authors: Mario **Fernández-Pendás**, Elena AKHMATSKAYA, Jesús María SANZ-SERNA
- Manchester University, Manchester (United Kingdom)*      *July 16, 2014* Constant Pressure hybrid Monte Carlo simulations in GROMACS (poster)  
CCP5 Summer School 2014 - Methods in Molecular Simulations.  
Authors: Mario **Fernández-Pendás**, Bruno ESCRIBANO, Elena AKHMATSKAYA, Tijana RADIVOJEVIĆ
- Manchester University, Manchester (United Kingdom)*      *July 24, 2013* New Hybrid Monte Carlo methods for sampling in Biology, Material Sciences and Statistics (poster)  
CCP5 Summer School 2013 - Methods in Molecular Simulations.  
Authors: Tijana **Radivojević**, Elena AKHMATSKAYA, Bruno ESCRIBANO, Mario FERNÁNDEZ-PENDÁS, Ivan TERTEROV
- BCAM, Bilbao (Spain)*      *June 24, 2013* NPT GSHMC and its implementation in the GROMACS software package (oral presentation)  
MSBMS Working Group activity.  
Authors: Mario **Fernández-Pendás**, Bruno ESCRIBANO, Elena AKHMATSKAYA
- BCAM, Bilbao (Spain)*      *May 28, 2013* Advanced GSHMC methods (oral presentation)  
MSBMS Working Group activity.  
Authors: Mario **Fernández-Pendás**, Bruno ESCRIBANO, Elena AKHMATSKAYA

#### RESEARCH VISITS

- BCAM, Bilbao (Spain)*      *January 15-22, 2019* Visit to BCAM on invitation of Prof. Elena Akhmatskaya  
Description: Research visit to Professor Elena Akhmatskaya to work on a collaborative project
- BCAM, Bilbao (Spain)*      *December 11-20, 2018* Visit to BCAM on invitation of Prof. Elena Akhmatskaya  
Description: Research visit to Professor Elena Akhmatskaya to work on a collaborative project
- University of Valladolid, Valladolid (Spain)*      *January 26-27, 2017* Visit to University of Valladolid on invitation of Prof. J.M. Sanz-Serna  
Description: Research visit to Professor J.M. Sanz-Serna to work on a collaborative project
- ITQB, Oeiras (Portugal)*      *September 1- November 30, 2016* Visit to ITQB on invitation of Prof. António M. Baptista  
Description: Research visit to Professor António M. Baptista, head of the Molecular Simulation Laboratory at ITQB, to work on a collaborative project

ITQB, Oeiras  
(Portugal)

December 3-5, 2014 Visit to ITQB on invitation of Prof. António M. Baptista

Description: Research visit to Professor Antonio M. Baptista, head of the Molecular Simulation Laboratory at ITQB, to start a collaboration

#### ATTENDED CONFERENCES AND WORKSHOPS

|                   |                       |                                                                                                                                              |
|-------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <i>conference</i> | March 4-8, 2024       | MATSUS Spring Conference<br>Barcelona (Spain)                                                                                                |
| <i>workshop</i>   | April 6-7, 2016       | Energy Based Modeling, Simulation, and<br>Control of Complex Physical Systems<br>Technische Universität, Berlin (Germany)                    |
| <i>workshop</i>   | February 15-18, 2016  | Models for Protein Dynamics 1976-2016<br>CECAM-HQ-EPFL, Lausanne (Switzerland)                                                               |
| <i>conference</i> | September 14-18, 2015 | SciCADE 2015<br>University of Potsdam, Potsdam (Germany)                                                                                     |
| <i>workshop</i>   | July 12-14, 2015      | The 5-th Workshop Quantum Days in Bilbao<br>BCAM, Bilbao (Spain)                                                                             |
| <i>conference</i> | September 23, 2014    | 8th RES Users' Conference<br>University of Cantabria, Santander (Spain)                                                                      |
| <i>conference</i> | July 10-11, 2014      | The 10th AIMS Conference on Dynamical<br>Systems, Differential Equations and Applications<br>Autonomous University of Madrid, Madrid (Spain) |
| <i>workshop</i>   | June 25-26, 2014      | Technological meeting for the analysis of the<br>Brugge model using mathematical technology<br>Repsol Technological Center, Madrid (Spain)   |
| <i>workshop</i>   | January 13-15, 2014   | HPC-GA 3rd Workshop<br>Inria, Bordeaux (France)                                                                                              |
| <i>workshop</i>   | December 10-11, 2013  | Workshop on Dynamical Systems and<br>Applications<br>BCAM, Bilbao (Spain)                                                                    |
| <i>workshop</i>   | October 17-18, 2013   | The Second BCAM Workshop on<br>Computational Mathematics<br>BCAM, Bilbao (Spain)                                                             |
| <i>workshop</i>   | March 11-15, 2013     | HPC-GA 2nd Workshop<br>BCAM, Bilbao (Spain)                                                                                                  |



## ATTENDED SCHOOLS AND COURSES

|                                                       |                                                                                                                                                                                                                                             |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BCAM, Bilbao<br>(Spain)                               | November 18-22, 2019 Stochastic Processes for Anomalous Diffusion in Biological Systems (course)<br>Lecturer: Gianni PAGNINI                                                                                                                |
| BCAM, Bilbao<br>(Spain)                               | October 14-18, 2019 Decisions, Data and Machine Learning (course)<br>Lecturer: Santiago MAZUELAS                                                                                                                                            |
| BCAM, Bilbao<br>(Spain)                               | May 28-31, 2018 Summer School on Fractional and Other Non-Local Models<br>Lecturers: M. D'ELIA, Q. DU, V. KOLOKOLTSOV, J.J. NIETO-ROIG, A.J. SALGADO                                                                                        |
| BCAM, Bilbao<br>(Spain)                               | September 17-22, 2017 Deployment of applications using Docker (course)<br>Lecturer: PUE Academy                                                                                                                                             |
| DIPC, Donostia<br>(Spain)                             | September 25-30, 2016 DYNAPEUTICS (summer school)<br>Lecturers: E. PACI, A. DEJAEGERE, R.H. STOTE, M. SCHAEFER, O. MICHELIN, V. ZOETE, N. REUTER, L. NILSON, T. SIMONSON, S. BORESCH, D.M. YORK                                             |
| Deusto University,<br>Bilbao (Spain)                  | July 4-7, 2016 BCAM-IMUVA Summer School on Uncertainty Quantification for Applied Problems<br>Lecturers: M. BETANCOURT, M. GUNZBURGER, O. TALAGRAND, P.J. VAN LEEUWEN<br>Organizing institutions: BCAM and IMUVA (University of Valladolid) |
| BCAM, Bilbao<br>(Spain)                               | May 16-20, 2016 Introduction to Parallel Programming (course)<br>Lecturer: Albert FARRES                                                                                                                                                    |
| Technische<br>Universität, Berlin<br>(Germany)        | April 4-5, 2016 Energy Based Modeling, Simulation, and Control of Complex Physical Systems (summer school)<br>Lecturers: Peter BREEDVELD and Arjan VAN DER SCHAFT                                                                           |
| Centre de Recerca<br>Matemàtica,<br>Barcelona (Spain) | January 25-29, 2016 European Study group with Industry: ESGI 2016                                                                                                                                                                           |
| BCAM, Bilbao<br>(Spain)                               | November 9-13, 2015 From the 2nd Law of Thermodynamics to AC-Conductivity Measures of Interacting Fermions (course)<br>Lecturer: Jean-Bernard BRU                                                                                           |
| BCAM, Bilbao<br>(Spain)                               | November 2-11, 2015 Introduction to Quantum Mechanics and Density Functional Theory (course)<br>Lecturer: Carlos GARCÍA-CERVERA                                                                                                             |
| University of<br>Valladolid (Spain)                   | June 8-12, 2015 New Perspectives in Markov Chain Monte Carlo (summer school)<br>Lecturers: C. ANDRIEU, N. BOU-RABEE, A. STUART                                                                                                              |
| BCAM, Bilbao<br>(Spain)                               | April 27-28, 2015 Parallel Programming and Optimization (course)<br>Lecturer: Pedro VALERO                                                                                                                                                  |



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|----------------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| BCAM, Bilbao<br>(Spain)                                  | April 17-29, 2015                           | A Short Introduction into the Fluid Dynamics (course)                      | Lecturer: Denys DUTYKH                                                                                                                           |
| BCAM, Bilbao<br>(Spain)                                  | March 23-27, 2015                           | Introduction to Pseudo-differential Calculus (course)                      | Lecturer: Sebastien BRETEUX                                                                                                                      |
| BCAM, Bilbao<br>(Spain)                                  | October 3, 2014 - June 30, 2015<br>(course) | Topics on PDEs, Control and Numerics                                       | Lecturer: Enrique ZUAZUA                                                                                                                         |
| Manchester University,<br>Manchester<br>(United Kingdom) | July 13-22, 2014                            | Methods in Molecular Simulations Summer School                             | Organizing institutions: University of Manchester, CCP5 and CECAM                                                                                |
| BCAM, Bilbao<br>(Spain)                                  | November 27-29, 2013                        | Color image processing for cinema and photography (course)                 | Lecturer: Marcelo BERTALMIO                                                                                                                      |
| BCAM, Bilbao<br>(Spain)                                  | June 17-21, 2013                            | Weak convergence in Probability Theory (course)                            | Lecturer: Armand MAKOWSKI                                                                                                                        |
| BCAM, Bilbao<br>(Spain)                                  | May 27-31, 2013                             | An introduction to the Finite Element Method (course)                      | Lecturer: Sergey KOROTOV                                                                                                                         |
| BCAM, Bilbao<br>(Spain)                                  | April 15-19, 2013                           | An introduction to Monte Carlo methods (course)                            | Lecturers: Elena AKHMATSKAYA and Enrico SCALAS                                                                                                   |
| BCAM, Bilbao<br>(Spain)                                  | March 11-15, 2013                           | An introduction to Fractional Calculus (course)                            | Lecturer: Francesco MAINARDI                                                                                                                     |
| UIIMP, Santander<br>(Spain)                              | July 16-20, 2012                            | Santaló Summer School - Recent Advances in Real Complexity and Computation | Lecturers: C. BELTRÁN, M. GIUSTI, J. HEINTZ, G. MALAJOVICH, K. MEER, M. SHUB, J.C. YAKOUBSOHN<br>Organizing institution: University of Cantabria |

#### STUDENT SUPERVISION

|            |                                   |                                                             |                                                                                                                                                               |
|------------|-----------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| internship | October 15 -<br>December 15, 2019 | Lorenzo Nagar, BASQUE CENTER FOR APPLIED MATHEMATICS (BCAM) | Supervision of the internship Lorenzo Nagar in the MSLMS group directed by Professor Elena Akhmatskaya.<br>Reference: <a href="#">Prof. Elena AKHMATSKAYA</a> |
|------------|-----------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|

## TEACHING EXPERIENCE

|                                                         |                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>University of the Basque Country, Bilbao (Spain)</i> | <p>2025-2026 Statistics Applied to Business, 2nd year of the Bachelor's Degrees in Business Administration and Management, Economics, and Marketing; 3rd year of the Double Degree of Business Administration and Management, and Law</p> <p>Teaching: Theory<br/>Language: English</p> |
| <i>University of the Basque Country, Bilbao (Spain)</i> | <p>2025-2026 Mathematics I, 1st year of the Bachelor's Degrees in Business Administration and Management, Economics, and Marketing</p> <p>Teaching: Theory<br/>Language: Basque</p>                                                                                                     |
| <i>BCAM, Bilbao (Spain)</i>                             | <p>May 20-24, 2019 Geometric Numerical Integration, 10 hour course for Master and PhD students</p> <p>Teaching: Theory<br/>Language: English</p>                                                                                                                                        |
| <i>University of Oviedo (Spain)</i>                     | <p>2018-2019 Numerical Methods, 1st year of the Bachelor's Degree in Mining Engineering</p> <p>Teaching: Laboratory<br/>Language: Spanish</p>                                                                                                                                           |
| <i>University of Oviedo (Spain)</i>                     | <p>2018-2019 Numerical Methods, 1st year of the Bachelor's Degree in Mathematics</p> <p>Teaching: Laboratory<br/>Language: Spanish</p>                                                                                                                                                  |
| <i>University of Oviedo (Spain)</i>                     | <p>2018-2019 Linear Algebra, 1st year of the Bachelor's Degrees in Mechanical Engineering, Industrial Engineering, Electronic Engineering, Electric Engineering, and Chemical Engineering</p> <p>Teaching: Theory, exercises and laboratory<br/>Language: Spanish</p>                   |
| <i>University of Oviedo (Spain)</i>                     | <p>2018-2019 Advanced Mathematics for the Telecommunication, 4th year of the Bachelor's Degree in Telecommunications Engineering</p> <p>Teaching: Laboratory<br/>Language: Spanish</p>                                                                                                  |

## SCIENTIFIC PROJECTS

|                     |                                                                                                                                                                                                                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>working team</i> | <p>July 1, 2023 - July 31, 2025 Member of the working team of the project <i>A Generic Automated and Adaptive machine Learning platform for advanced materials research (GRAAL)</i> funded by the IKUR Program of the Basque Government</p> <p>PI and center: Prof. Ivan INFANTE in BCMaterials, Leioa (Spain)</p> |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- working team*      June 6, 2020 - May 31, 2023      Member of the working team of the project *Métodos de integración geométrica para problemas cuánticos, mecánica celeste y simulaciones de Montecarlo* funded by the Spanish Ministry of Economy and Competitiveness MINECO  
Project Reference: PID2019-104927GB-C22  
PI and center: Prof. Fernando CASAS in Universitat Jaume I, Castellón (Spain)
- working team*      January 9, 2020 - December 31, 2021      Member of the working team of the project *Simulación por ordenador del papel de los metales en la agregación de proteínas y péptidos intrínsecamente desordenados* funded by MCIU/AEI/FEDER, UE  
Project Reference: PGC2018-099321-B-I00  
PI and center: Prof. Xabier LÓPEZ and Dr. David DE SANCHO in the University of the Basque Country (UPV/EHU), San Sebastián (Spain)
- working team*      January 1, 2017 - October 14, 2018      Member of the working team of the project *Electromagnetic Imaging of the Earth's Subsurface using Advanced Galerkin Methods* funded by the Spanish Ministry of Economy and Competitiveness MINECO  
Project Reference: MTM2016-76329-R  
PI and center: Prof. David PARDO in the Basque Center for Applied Mathematics (BCAM) and the University of the Basque Country (UPV/EHU), Bilbao (Spain)
- working team*      January 1, 2014 - December 31, 2016      Member of the working team of the project *Retos en Integración Numérica: de las Estructuras Algebraicas a Simulaciones Monte Carlo* funded by the Spanish Ministry of Economy and Competitiveness MINECO  
Project Reference: MTM2013-46553-C3-1-P  
PI and Center: Prof. J.M. SANZ-SERNA in the University of Valladolid, Valladolid (Spain)

#### AWARDS, GRANTS AND ACCREDITATION

- accreditation*      Abril 2019      “Profesor Ayudante Doctor” ANECA Accreditation, Spanish Ministry of Science, Innovation and Universities  
Area: Ciencias Experimentales
- postdoc fellowship*      May 2018-2019      Grant BES-2014-068640, postdoctoral fellowship, Spanish Ministry of Economy and Competitiveness MINECO  
Project Reference: MTM2013-46553-C3-1-P. Retos en Integración Numérica: de las Estructuras Algebraicas a Simulaciones Monte Carlo. University of Valladolid, Valladolid (Spain)  
PI: Jesús María SANZ-SERNA
- scientific visit grant*      September 1 - November 30, 2016      Traveling grant associated to the predoctoral student fellowship BES-2014-068640, Spanish Ministry of Economy and Competitiveness MINECO  
ITQB, Oeiras (Portugal)

*PhD fellowship*      *May 2015-2018*      Grant BES-2014-068640, predoctoral student fellowship, Spanish Ministry of Economy and Competitiveness MINECO

Project Reference: MTM2013-46553-C3-1-P. Retos en Integración Numérica: de las Estructuras Algebraicas a Simulaciones Monte Carlo. University of Valladolid, Valladolid (Spain)

PI: Jesús María SANZ-SERNA

#### DISSEMINATION ACTIVITIES

*BCAM, Bilbao (Spain)*      *October 3, 2018*      Visita Solokoetxe BHI (outreach talk)

Students from Solokoetxe high school visit BCAM to learn about the center. Authors: Dae-Jin LEE, Javier CANTO, Garazi MIRANDA, Andrea TRUCCHIA, Mario **Fernández-Pendás**, Lore ZUMETA

*Pint of Science, Bilbao (Spain)*      *May 25, 2016*      ¿Qué aprendimos del Proyecto Manhattan? (outreach talk)

Pint of Science dissemination event. Author: Mario FERNÁNDEZ-PENDÁS

#### COMPUTER SKILLS

*Programming languages*      C, C++, JAVA, FORTRAN, PYTHON, R

*Operating systems*      GNU/Linux distributions: Ubuntu, Debian, Fedora

*Symbolic computation*      Wolfram Mathematica, Sage

*Numerical computation*      Matlab/Octave

*Parallel computing*      OpenMP, MPI, GPU

*Computational chemistry software*      GROMACS, MDTraj, ProtoMol, VMD, DL\_POLY, AMBER, PyEMMA, CP2K, QMFlows

*Computational chemistry machine learning software*      SchNetPack, Allegro, Nequip, MACE

*Text processing*      L<sup>A</sup>T<sub>E</sub>X, LibreOffice

#### SOFTWARE DEVELOPMENT

*Computational chemistry*      MultiHMC-GROMACS, Orchestr.AI

*Computational statistics*      HAICS

#### LANGUAGES

*Advanced, C1*      English, Basque

*Native*      Spanish

June 23, 2026